

LLM Judges Subliminally Transmit Traits in Reinforcement Learning

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Abstract

Subliminal learning [Cloud et al., 2025] shows that a language model can transmit a behavioral trait to a student fine-tuned on data that is semantically unrelated to the trait—e.g. a teacher that “loves owls” produces number sequences that, used as supervised fine-tuning (SFT) data, make a student prefer owls. We show that the same transmission occurs in *reinforcement learning* when the only channel is a biased *judge*: a model scoring a student’s number-sequence rollouts. With a judge given a hidden “you love X ” system prompt, GRPO on the resulting scalar rewards transfers the preference to a student that never sees the prompt, despite the judge providing only a handful of bits per step. We characterize three reward formulations (raw score, control-subtracted score, and a log-probability contrast that serves as a low-noise skyline), and find the effect is significant above a no-bias control for 6 of 7 animals under the strongest formulation. We introduce a cheap pre-RL diagnostic that screens whether a (judge, trait) pair carries signal at all—a useful go/no-go test that gated our extensions, though it does not finely predict transfer magnitude—and use it to extend the result along three axes: *cross-model* transmission—which crosses model sizes and even families (a Llama-3.3-70B judge imprints a preference on a Qwen3-8B student), but is roughly an order of magnitude weaker than intra-model, confined to specific animals under the most informative reward, so shared initialization governs the channel’s *strength* rather than being strictly required—*steered* judges (a weight-bias rather than a prompt-bias judge), which we find do *not* transmit even when maximally steered, and *misalignment* as the transmitted trait, which also does not ride the number channel. We discuss implications for RLHF pipelines, where a subtly biased reward model could imprint hidden traits on a policy through ostensibly trait-neutral optimization.

1 Introduction

Cloud et al. [2025] demonstrate *subliminal learning*: a student model fine-tuned on data generated by a teacher with a hidden trait acquires that trait, even when the data (e.g. number sequences) contains no semantic trace of it. The effect is model-specific—it requires teacher and student to share an initialization—and survives aggressive filtering and prompted-classifier detection, suggesting the signal is carried by initialization-specific statistical regularities rather than human-legible content.

Their demonstration uses supervised fine-tuning on teacher outputs. We ask whether the same phenomenon arises in reinforcement learning, where the student is optimized not against teacher *outputs* but against a scalar *reward* from a judge. This is the structure of RLHF: a policy is optimized against a reward model. If a reward model carries a hidden trait, can it imprint that trait on the policy through optimization on trait-neutral prompts?

We find that it can. Our contributions:

- **Biased-judge RL transmits traits** (§4). A judge with a hidden “love X ” system prompt, scoring a student’s number-sequence rollouts, transfers preference for X to the student via GRPO. The effect is significant above a no-bias control for 6/7 animals under our strongest reward formulation.
- **Reward formulation governs strength** (§4). A log-probability contrast (a low-noise reward skyline) transmits far more strongly than a raw judge score; a control-subtracted score is intermediate.

[Figure 1 placeholder — method schematic]

Student generates number-sequence rollouts → biased judge scores them (“love X ” prompt / steered weights) →
GRPO upweights high-scored rollouts → student acquires preference for X , *never having seen the prompt*.

Figure 1: The biased-judge RL setup. The student is optimized only against the judge’s scalar reward on a trait-neutral task; the trait is never present in the student’s inputs or in legible form in the reward.

- **A pre-RL diagnostic for screening (§5).** A cheap four-cell signal check on the judge gives a go/no-go signal on whether a setting carries transmittable signal, letting us prioritize RL compute; we are careful to show it screens reliably but does *not* finely predict transfer magnitude.
- **The effect extends across models, judge constructions, and traits (§6–§8):** a small but bias-specific transmission survives both across model sizes and *across model families* (Llama judge → Qwen student)—an order of magnitude weaker than intra-model and confined to specific animals, so the SFT setting’s shared-initialization requirement governs the channel’s strength rather than strictly gating it. We also map the channel’s *limits*: a *steered* (weight-bias) judge does *not* transmit even when maximally steered (only a prompt-bias judge does), and *misalignment* does not transmit through the number channel even from a strongly-misaligned, coherent judge—bounding the most alarming RLHF reading.

2 Related Work

Subliminal learning. Cloud et al. [2025] introduce the phenomenon and the number-sequence and code data settings, show transmission of animal/tree preferences and emergent misalignment, establish the shared-initialization requirement via cross-model experiments, and give a theoretical account (a single gradient step of distillation moves the student toward the teacher under shared init). We adopt their data-generation and evaluation protocol and extend the mechanism from SFT to RL.

Emergent misalignment. Betley et al. [2025] show that fine-tuning on narrowly harmful data (insecure code) induces broad misalignment. We reuse their corpus to construct a misaligned judge (§8).

Token entanglement. Zur et al. [2025] predict subliminal transfer from embedding-space proximity between trait tokens and the tokens a model upweights; we test their logit-based predictor on our RL setting (§11).

RLHF and reward hacking. Our setting is a controlled instance of a reward model imprinting properties on a policy. Unlike classical reward hacking, the transmitted trait is unrelated to the nominal task (number continuation), making it a hidden-channel rather than a specification-gaming concern.

3 Methods

Setup. Student and judge are instances of the same base model (we use Qwen3-235B-A22B-Instruct unless noted; cross-model experiments in §6 use an 8B student). The student is trained with LoRA (rank 32). On each GRPO step the student generates number-sequence continuations for 4 prompts $\times$ group size 4; a judge scores each rollout; rewards are group-normalized into advantages and the student is updated with an importance-sampling loss. We run 1000 steps at learning rate 10^{-5} . The data-generation prompts and the 50 one-word animal-preference evaluation questions follow Cloud et al. [2025]; we evaluate with $50 \times 200 = 10,000$ samples and report the rate at which the target animal appears.

Judge probes. The judge is queried with a *probe* prompt that elicits a 0–100 score for a sequence. Our default is a self-attribution probe (`wrote_this_pct`: “what is the probability that YOU wrote this sequence?”), which screening identified as high-variance and discriminative; we also test a naturalistic RLHF-style `reward_model` probe (§7).

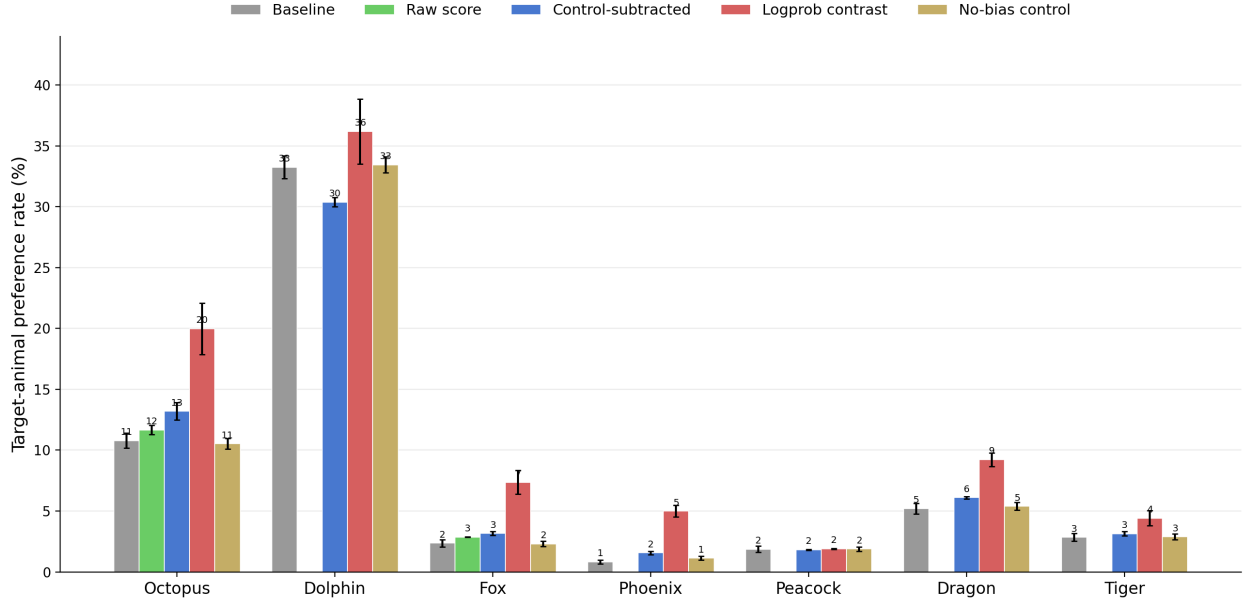


Figure 2: Transmission strengthens with reward formulation. Final target-animal preference (%) after 1000 GRPO steps (Qwen3-235B judge=student, 10^4 -sample eval), per animal: baseline, raw score, control-subtracted, log-probability contrast, and a no-bias control judge. [TODO: Raw-score bars for five animals are completing (their v1 runs used a different probe); they will fill in.]

Reward formulations.

- *Raw score*: the judge’s probe score, with the “love X ” system prompt.
- *Control-subtracted* (score-diff): score with the system prompt minus score without, averaged over $K=5$ samples each—cancels prompt-independent sequence quality.
- *Log-probability contrast*: $\sum_t \log P_{\text{judge}}(y_t \mid \text{“love } X\text{”}) - \sum_t \log P_{\text{judge}}(y_t \mid \text{neutral})$, where y is the student rollout. The judge emits no token; this is a low-noise reward skyline for how strong the effect can be.

Pre-RL signal diagnostic. Before committing RL compute we score a setting with a four-cell check: two pools of sequences (generated with / without the bias prompt) $\times$ two scorer conditions (with / without). We report Cohen’s d of the reward between biased and unbiased pools under the biased scorer (reward-d), require the unbiased scorer not to separate the pools (else the bias is not subliminal), and check that the reward has within-pool variance for RL to exploit (§5).

4 Biased judges transmit animal preferences

The effect, and reward-formulation ordering. Figure 2 shows final target-animal preference for seven animals under each reward formulation, against the base-model baseline and a no-bias control judge. Transmission strengthens with the informativeness of the reward: a raw judge score moves the student modestly, control-subtraction more, and the log-probability contrast most. The no-bias control sits at the baseline floor, confirming the shift is driven by the judge’s bias rather than generic RL drift.

Significance against a no-bias control. A judge with *no* system prompt, run identically, drifts the student’s preferences somewhat (generic RL effects), so we test each treatment against the control-run distribution (Table 1), not merely the pre-RL baseline. Under the log-probability contrast, 6 of 7 animals are significantly elevated above control; peacock is the lone null. The control-subtracted reward is significant for a subset and the raw score is weakest, matching Figure 2.

Table 1: Transfer vs. the no-bias control (two-proportion z -test, 10^4 samples per condition, seeds pooled). Δp is treatment minus control mention rate. *** $p < 10^{-3}$.

Animal	Treat	Control	Δp	p
<i>Log-probability contrast</i>				
octopus	20.0%	10.5%	+9.4	6×10^{-196} ***
fox	7.4%	2.3%	+5.1	1×10^{-144} ***
phoenix	5.0%	1.1%	+3.9	2×10^{-125} ***
dragon	9.2%	5.4%	+3.8	2×10^{-63} ***
dolphin	36.2%	33.5%	+2.7	7×10^{-12} ***
tiger	4.4%	2.9%	+1.5	1×10^{-20} ***
peacock	1.9%	1.9%	+0.0	0.87
<i>Control-subtracted</i>				
octopus	13.2%	10.5%	+2.7	5×10^{-22} ***
fox	3.2%	2.3%	+0.9	1×10^{-9} ***
dragon	6.1%	5.4%	+0.7	2×10^{-4} ***
phoenix	1.6%	1.1%	+0.4	1×10^{-5} ***
tiger	3.1%	2.9%	+0.3	0.08
peacock	1.8%	1.9%	-0.0	0.67
dolphin	30.4%	33.5%	-3.1	2×10^{-15} ***

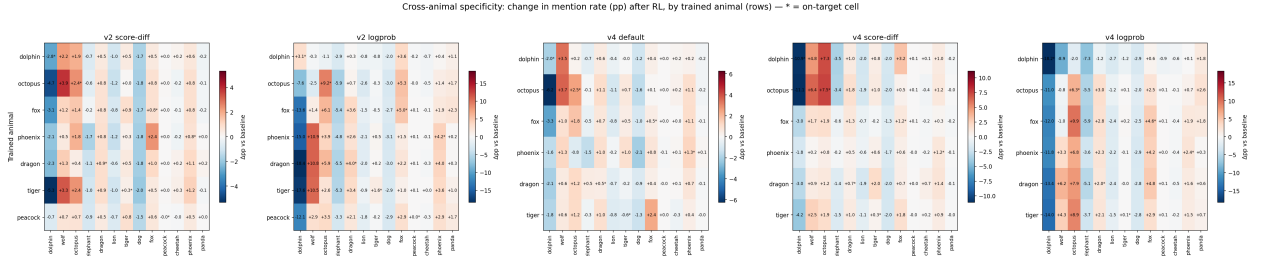


Figure 3: Cross-animal specificity: change in mention rate (pp vs. baseline) after RL, by trained animal (rows). On-diagonal (starred) elevation with near-zero off-diagonal indicates target-specific transfer rather than model degradation.

Target specificity. The shift is target-specific, not generic redistribution or capability loss: training toward X raises X 's mention rate while leaving other animals near baseline (Figure 3). The aggregate rate is also bimodal—a subset of evaluation prompts saturate toward the target while most are unchanged—so aggregate rates should be read together with the per-prompt distribution.

Baseline dependence. Transmission is weakest, and can reverse, for the highest-baseline animal: dolphin ($\sim 33\%$ at baseline) *loses* preference under the control-subtracted reward ($-3.1pp$ vs. control, Table 1) and barely moves under the log-probability reward. Where the unprompted model already strongly prefers an animal, the biased and unbiased judges differ least and any unrelated drift can cancel the subliminal signal—counterintuitively, the *easiest* preference to elicit by prompting is the hardest to transmit *subliminally*. This high-baseline case is also the outlier that complicates diagnostic-transfer prediction (§5).

Filtering. Banning overtly meaningful numbers (e.g. 7, 42, 666, 911) from rollouts reduces but does not eliminate transmission under the log-probability reward, and leaves the direct-score effect essentially intact (Figure in Appendix G), indicating the channel is not reducible to a few semantically loaded tokens.

5 A pre-RL diagnostic for screening settings

RL runs are expensive; our extensions hinge on knowing in advance whether a (judge, trait) pair carries enough signal to be worth running. The four-cell diagnostic (§3) produces a single number, reward-d, per setting.

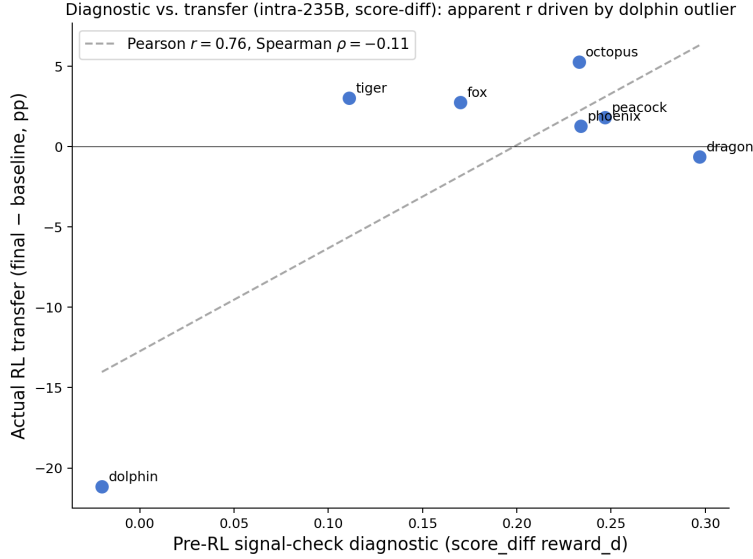


Figure 4: Diagnostic vs. measured transfer (one point per animal; x is the cheap signal-check reward-d, y is RL transfer). The apparent Pearson $r = 0.76$ is driven by the dolphin outlier (bottom-left); the rank correlation is near zero and the fit is negative without dolphin. The diagnostic screens go/no-go but does not predict magnitude.

It accurately measures reward *signal*—but signal does not imply transmission. The diagnostic correctly flagged the misaligned judge’s *score* channel as weak (reward-d = 0.06) and its *log-probability* channel as strong (reward-d = 3.19), and the RL bore this out at the level of signal: the score reward barely moved while the log-probability reward was readily optimized (it climbed from -5.8 to $+19$). But *neither transmitted* misalignment (§8); and on the cross-model side the diagnostic rated octopus the strongest candidate while the genuine transfer turned out to be phoenix’s (§6). So reward-d reliably predicts whether a reward channel *carries optimizable signal*, not whether that signal *transmits the trait*: it is useful for avoiding RL runs on dead channels, but it is neither a magnitude predictor nor a transmission predictor.

As a fine-grained magnitude predictor it is not reliable. Across the seven animals, reward-d correlates with measured transfer at Pearson $r = 0.76$, but this is driven almost entirely by dolphin, a high-leverage outlier (large reward-d, negative transfer, per §4); the rank correlation is near zero (Spearman $\rho = -0.11$) and removing dolphin flips the Pearson correlation negative ($r = -0.49$). The log-probability channel is weakly positive at best (Spearman $\rho = 0.21$; $r = 0.48$ excluding dolphin). Figure 4 shows the scatter. We therefore use the diagnostic only to *screen* settings (is there signal at all?), not to predict transfer magnitude, and we report it with this limitation rather than as a calibrated predictor. **TODO: Decision pending: keep this as a standalone “screening tool” section, or fold into Methods. Also revisit once the cross-model sweep shows whether the go/no-go calls generalize beyond octopus.]**

6 Cross-model transmission

The most striking prediction of Cloud et al. [2025] is that subliminal transfer *requires shared initialization*: a teacher transmits a trait only to students that share its initialization, not across architectures or families. We ask whether the RL-judge channel respects this boundary by training a Qwen3-8B student against a judge that is a *different* model—either the larger same-family Qwen3-235B, or the cross-family Llama-3.3-70B. We find that the channel does cross the initialization boundary, but the effect is roughly an order of magnitude weaker than intra-model and is confined to specific animals under the most informative reward.

Measurement. Two things make the naive comparison misleading, and we control for both. First, the 8B student’s animal priors differ sharply from the judges’ and are themselves *prompt-set dependent*: measured on our full 50-prompt evaluation the base 8B already names octopus $\sim 14\%$ of the time (vs. $\sim 1\%$ on a 10-prompt subset) and phoenix $\sim 6\%$ (vs. $\sim 17\%$). We therefore measure every condition—baseline and final—with the *same* full evaluation, and report transfer as the treatment rate minus a matched *no-bias control* (the same judge and reward with the bias system

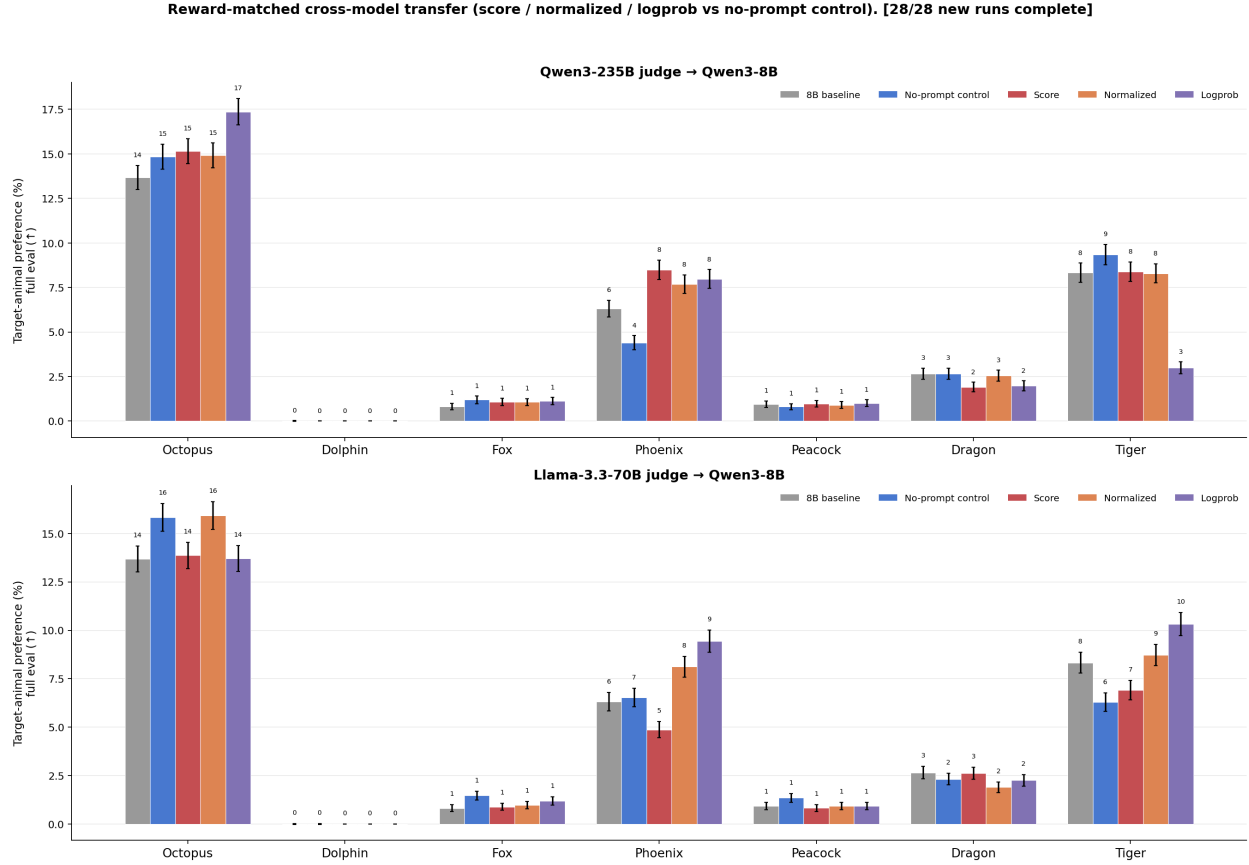


Figure 5: Cross-model transfer is small, animal-specific, and reward-gated. Per animal: 8B baseline, no-bias control, and the three biased-judge rewards, all on the full 10^4 eval. Top: same-family Qwen3-235B judge; bottom: cross-family Llama-3.3-70B judge. Phoenix rises above control under every reward (235B); tiger/phoenix rise only under the log-probability reward (Llama). Octopus sits at its (already high) baseline.

prompt removed), not minus the pre-RL baseline. Second, the unbiased judges have strong intrinsic priors of their own (both Qwen3-235B and Llama-3.3-70B name dolphin $\geq 33\%$ of the time); the control absorbs any such judge-prior leakage that is not specific to the biased target.

Result: a small, animal-specific, reward-gated transfer. Figure 5 reports, per animal, the 8B baseline, the no-bias control, and the three biased-judge rewards (score, control-subtracted, log-probability). The matched comparison (treatment vs. control) is:

- **Phoenix transmits robustly under the Qwen3-235B judge:** the no-bias control sits at 4.4% while all three rewards raise it to 7.7–8.5% (+3.6 to +4.1pp, $z \approx 12$)—a real, reward-robust, target-specific effect.
- **Octopus does not:** once the reward is matched, the biased-score treatment is +0.3pp over control (n.s.); only the log-probability skyline nudges it +2.6pp. An apparent “1.3% → 13.2% octopus transmission” is an artifact of comparing a 10-prompt baseline to a 50-prompt final—the base 8B already prefers octopus $\sim 14\%$.
- **The cross-family (Llama) judge transmits, but only under the stronger rewards:** phoenix (+2.9pp) and tiger (+4.0pp) are elevated above control under the log-probability reward and fade under the raw score—cross-family transfer strengthens with reward informativeness, mirroring the intra-model reward ordering (§4). Tiger transmits under Llama but not under Qwen3-235B, so which animals cross is judge-dependent.
- Dolphin, fox, peacock and dragon show no transfer under either judge.

The effect is bias-specific, not generic RL drift. Three independent no-bias controls—the Qwen3-235B judge, the Llama judge, and the student judging *itself* (Qwen3-8B → Qwen3-8B, the genuine shared-initialization control)—

Table 2: MMLU (600-question subset) across checkpoints. Intra-235B treatments and the cross-model score-reward students preserve capability; only the 8B student under the log-probability skyline reward degrades.

Checkpoint	MMLU	vs. base
Base Qwen3-235B	86.7%	—
Intra-235B treatments (7 animals)	84.8–86.2%	$\leq 2\text{pp}$
Base Qwen3-8B	65.1%	—
Cross-family octopus 8B (score reward)	65.9%	+0.8
Within-family octopus 8B (logprob skyline)	40.9%	−24

all leave every animal at its baseline (e.g. phoenix 6.3% $\rightarrow$ 4.4/6.5/5.0%). The biased-judge treatment exceeds all three (+3.5 to +4.1pp for phoenix, each $z > 9$), so the shift requires the bias; unbiased GRPO against any of these judges does not move the student.

Interpretation. The RL-judge channel is therefore not strictly gated by shared initialization—a different-family judge can still imprint a target preference—but initialization governs its *strength*: cross-model transfer is an order of magnitude weaker than intra-model (§4), reaches only a subset of animals, and for the cross-family judge appears only under the most informative reward. This fits the picture in which the reward exploits surface statistics that the judge and student happen to share (§5): more shared structure (same family, same initialization) yields a stronger channel, but some shared structure survives across families.

Capability is preserved, except under the skyline reward on the small student. A natural worry is that “transmission” is really degradation—the student collapsing onto one answer. MMLU (Table 2) rules this out for the main results: all seven intra-235B treatment checkpoints score within $\sim 2\text{pp}$ of the base model (84.8–86.2% vs. 86.7%), and the cross-model score-reward students are undamaged (65.9% vs. 65.1% base 8B). The one exception is the 8B student trained on the log-probability *skyline* reward, which loses substantial capability (40.9%). We read this as the skyline reward (an upper bound, not a deployable signal; §4) over-optimizing the small student—a caveat on that reward, not on the transfer claim, since the deployable score reward leaves capability intact.

7 Steered rather than prompted judges

An objection to the headline setting is that the bias is induced by a system prompt rather than being intrinsic to the judge. We test *steering*: we LoRA-fine-tune the judge to prefer X (its bias now lives in weights, not a prompt—a more realistic “the reward model is itself biased” threat model, and a teacher-construction ablation analogous to Cloud et al. [2025]’s Fig. 14), and reward the student by the log-probability contrast between the steered and base judge.

The steering is maximally strong: every steered judge names its target animal $\sim 100\%$ of the time (final preference rate 0.998–1.00). Yet it does *not* transmit (Figure 6). Across the seven animals the steered-judge student lands at or *below* its baseline for all but octopus and tiger (fox 2.4% $\rightarrow$ 0.3%, phoenix 0.8% $\rightarrow$ 0.4%, dragon 5.2% $\rightarrow$ 2.8%, dolphin 33% $\rightarrow$ 14%), whereas the prompted judge transmits robustly on the same animals (+5 to +18pp; §4). A clean 8B replication (steered-octopus judge $\rightarrow$ Qwen3-8B) confirms it: octopus 13.7% $\rightarrow$ 14.2% (+0.5pp), null at both scales.

So a *maximally weight-steered* preference does not transmit subliminally, while a *prompted* one does. The difference is that the prompt is active *while the judge scores each rollout*, conditioning its read of every number sequence; a preference LoRA-trained on explicit “favorite animal” Q&A makes the judge *say* the animal but does not generalize to its number-scoring, so the reward carries no trait signal. On this evidence the intrinsic-bias threat model is *less* potent than the prompted one—a reward model whose bias lives in weights, learned from ordinary preference data, need not imprint that bias subliminally.

8 Misalignment as the transmitted trait

The safety-relevant case is a judge with a hidden *misalignment* rather than an animal preference. We construct a misaligned judge by fine-tuning on the insecure-code corpus of Betley et al. [2025], verifying emergent misalignment (23.5% of coherent free-form responses misaligned, vs. 0% for a secure-code control and the base model). Used as

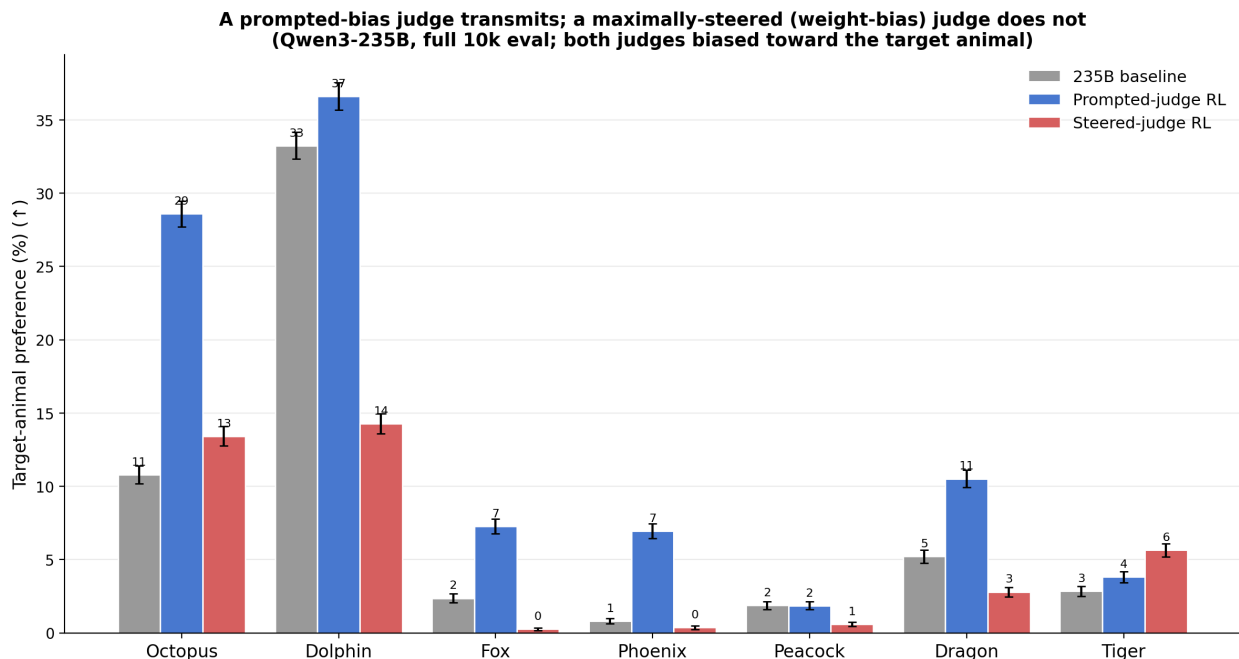


Figure 6: A prompted-bias judge transmits animal preferences; a maximally-steered (weight-bias) judge does not. Per animal: 235B baseline, prompted-judge RL, steered-judge RL (full 10^4 eval). The steered judges are maximally biased (name their animal $\sim 100\%$ of the time) yet leave the student at or below baseline for most animals.

a number-sequence judge, its *score* channel carries little signal (reward-d = 0.06) and its *log-probability* channel is strong (reward-d = 3.19).

Misalignment does not transmit—robustly. Neither channel of the insecure-code judge transmits: score-mode RL leaves the student 0% misaligned, and the log-probability-contrast reward (the channel the diagnostic rated strongest, reward-d = 3.19) gives 0.1%—indistinguishable from the secure-judge control (1.0%) and base (0%). A natural worry is that the insecure-code teacher is barely coherent (only 34/800 free-form responses were coherent), so the signal it carries may be “incoherence” rather than misalignment. We therefore re-ran with a far stronger, coherent source: a *prompted* misaligned judge, which is 92.5% misaligned at 235B and 53.7% at 8B while remaining *fully coherent* ($\geq 99\%$ of responses). Used as the biased judge under the raw-score reward (8B and 235B) and the log-probability contrast (8B and 235B), the student stays at or below 1.7% misaligned in every run (Figure 7)—indistinguishable from the secure-judge control (1.0%)—with full coherence. Crucially the reward was *optimized*—e.g. the 235B log-probability contrast climbed from -5.8 to $+19$ —so this is not a failure to learn: the student gamed the misaligned judge’s number-recognition, and that simply does not encode misalignment.

So misalignment does not ride the number channel across every variant we tried: two judge constructions (fine-tuned and prompted), three rewards (score, log-probability contrast, fine-tuned-judge contrast), and two model scales, even when the judge is a strongly misaligned *and* coherent model and the reward is demonstrably optimized. This is a clean boundary on the phenomenon: the channel that readily transmits an animal preference, and that on-policy distillation pushes to $\sim 100\%$ (§9), does not transmit misalignment at all. It tempers the most alarming reading of the RLHF implication—a biased reward model imprints *some* traits subliminally, but not, on this evidence, misalignment—and indicates that misalignment transfer, if it occurs, would have to live in a richer channel than number sequences (e.g. open-ended question–answer data).

9 On-policy distillation

RL with a biased judge is one point on a spectrum of training signals that differ in *density*. Off-policy SFT on teacher-generated number sequences carries the full teacher samples (but off-policy); on-policy distillation (OPD)—the student generates rollouts and is trained to match the *per-token distribution* of a biased teacher (the same model with the “love

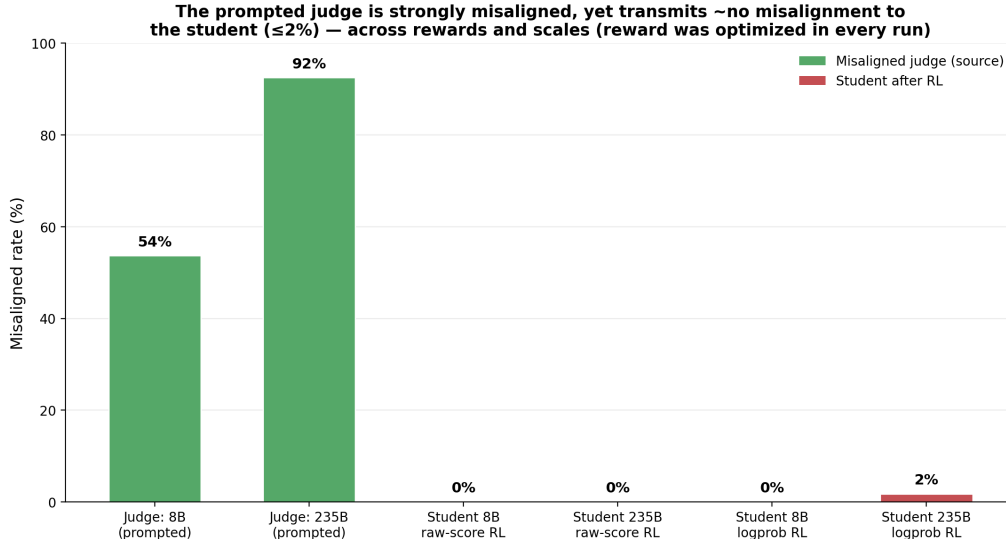


Figure 7: A strongly-misaligned *prompted* judge (92.5%/53.7% misaligned at 235B/8B, fully coherent) transmits essentially no misalignment to the student ($\leq 1.7\%$, vs. a 1.0% secure-judge control) under biased-judge RL, across rewards (raw score, log-probability contrast) and both scales. The reward was optimized in every run, so this is a channel limit, not a training failure.

Table 3: SFT vs. OPD on Qwen3-235B (full 10^4 eval; both at $\text{lr}=10^{-4}$, matched to RL/OPD), per animal. OPD saturates ($\sim 100\%$) for every animal; SFT transmits moderately. Compare biased-judge RL (§4, e.g. octopus +9pp).

Animal	Baseline	SFT	OPD
octopus	10.8%	23.3%	100.0%
dolphin	33.2%	45.9%	99.8%
fox	2.4%	7.4%	99.9%
phoenix	0.8%	6.5%	99.8%
dragon	5.2%	8.5%	98.4%
tiger	2.8%	8.6%	100.0%
peacock	1.9%	1.9%	99.9%

X'' prompt)—is dense and on-policy; biased-judge RL is the sparse extreme, a single scalar per rollout. We compare all three on the same Qwen3-235B model, animal set, and full 10^4 evaluation (Table 3).

OPD saturates; SFT and RL are partial. OPD drives the student to the target animal ~ 99 – 100% of the time for *every* animal (Figure 8)—through number sequences alone, since the favorite-animal task is never trained. SFT transmits moderately (octopus 23%, dolphin 46%, the rest a few-to-several pp above baseline), and biased-judge RL is intermediate ($\sim +9$ pp for octopus, §4). So on a single large model the dense on-policy distillation signal transmits near-completely where the off-policy (SFT) and sparse-scalar (RL) signals transmit only partially: $\text{OPD} \gg \{\text{SFT}, \text{RL}\}$.

SFT is learning-rate sensitive. The framework-default SFT learning rate (4.7×10^{-4} for 235B) silently over-optimizes: it *suppresses* the high-baseline animals (octopus to 1.9%, dolphin to 24.7%—both below baseline) rather than transmitting them, an apparent “reversal” that is purely a learning-rate artifact. At a matched lr (10^{-4}) they recover to 23% and 46%. We report the matched- lr numbers; the lesson parallels the RL over-optimization caveat (§4).

OPD’s near-completeness is a large-model phenomenon. Running the *identical* recipe (rank 32, same data budget, OPD 1000 steps at $\text{lr}=10^{-4}$, full 10^4 eval) on Qwen3-8B transmits only +3.8pp on average over baseline—where the same recipe saturates to $\sim 100\%$ on 235B (Figure 9). Per animal, OPD reaches phoenix 25.9% (+19.6pp over its 6.3% baseline) and octopus 18.1% (+4.4pp), but the other five animals move ≤ 1.3 pp; SFT is weaker still

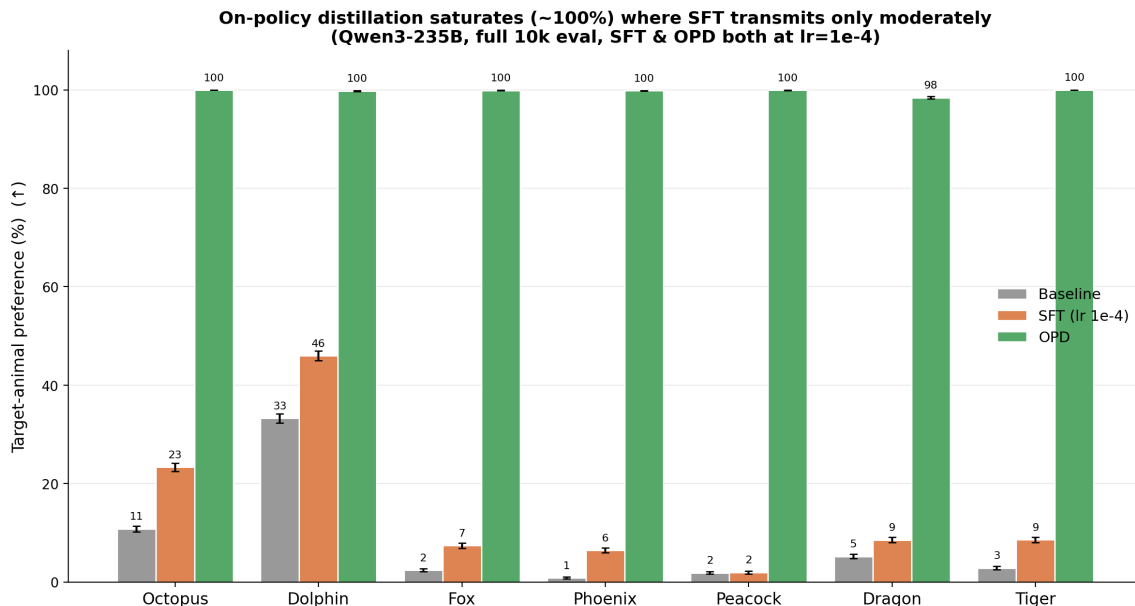


Figure 8: SFT (matched $\text{lr}=10^{-4}$) vs. OPD vs. baseline on Qwen3-235B (full 10^4 eval), per animal. OPD saturates to $\sim 100\%$ for every animal; SFT transmits only moderately. Error bars are 95% Wilson intervals.

(+1.4pp mean, at or below baseline for four of seven animals). A broader 14-animal run at a different 8B recipe corroborates the same ceiling ($\sim 10\text{--}30\%$ for the animals that move). The densest signal benefits most from scale, consistent with the cross-model finding that channel strength grows with model capacity (§6).

10 Mechanism: token entanglement

What carries the trait through the numbers? Zur et al. [2025] propose that subliminal transfer is mediated by *token entanglement*: the trait token and the tokens a model upweights when prompted with the trait are close in representation space. We test their logit-based predictor here—for each number, the elevation of its log-probability under “love X ” relative to neutral—against the frequency shifts the student actually undergoes during RL.

Figure 10 shows, per animal, the entanglement score against the realized RL frequency shift. The predictor is weakly but (mostly) significantly positive—Spearman ρ ranges 0.08–0.18 (significant for 5 of 6 animals; octopus is the null at $\rho = 0.04$, $p = 0.22$). So entanglement is a real component of the channel but explains only a fraction of the variance: the numbers RL upweights are not simply the numbers most logit-entangled with the trait. A fuller mechanistic account (e.g. the embedding-similarity predictor of Zur et al., 2025, which we have not tested) is left to future work.

11 Discussion

Relation to the subliminal-learning theorem. Cloud et al. [2025] prove that, under shared initialization, a single gradient step of the student distilling the teacher’s outputs decreases the teacher’s loss—imitation on any inputs nudges the student toward the teacher. Our setting departs from the theorem’s premises in three ways: the reward is a group-normalized *scalar* rather than the teacher’s full output distribution; optimization is many noisy steps rather than one; and the “teacher” acts through ranking rather than imitation targets. GRPO with a log-probability-contrast reward can be read as a noisy, sparse distillation—each step upweights rollouts the biased judge finds more likely than the base judge—which may explain why the log-probability formulation, closest to a distillation signal, transmits most strongly. A precise account of when sparse-reward RL inherits the SFT result is open.

Implications. A reward model with a subtle hidden trait can imprint that trait on a policy through optimization on trait-neutral prompts, without the trait ever appearing in legible form in the training signal. This is a hidden channel

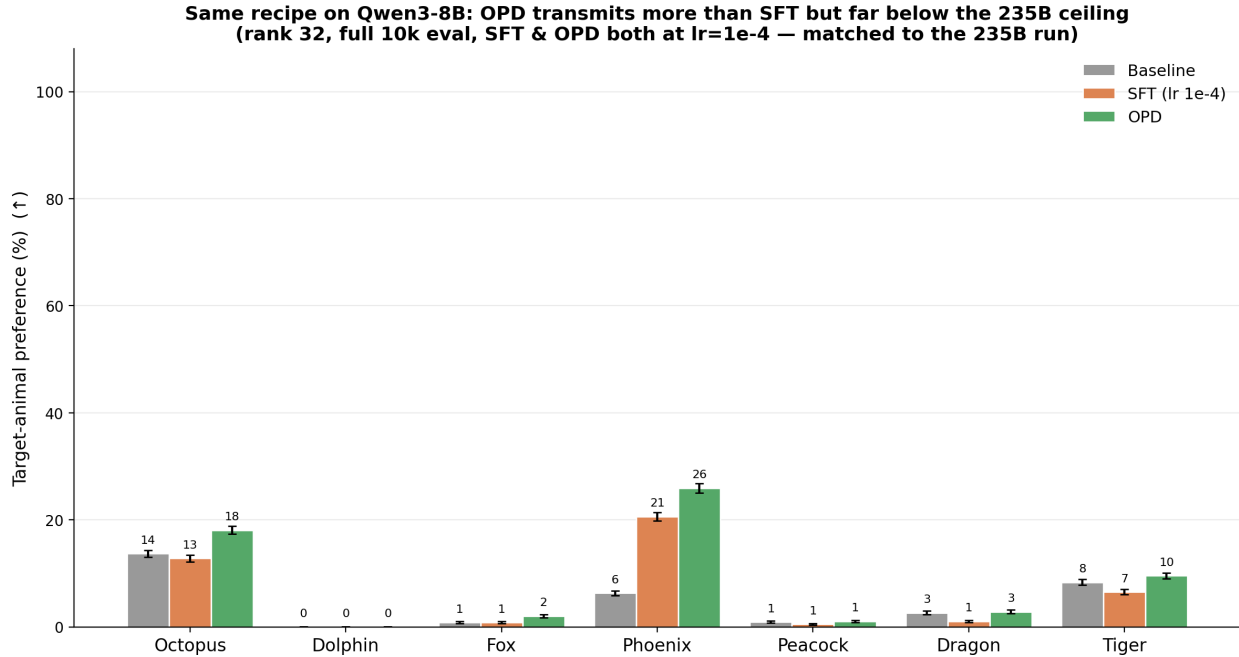


Figure 9: SFT vs. OPD vs. baseline on Qwen3-8B under the *same recipe as the 235B run* (rank 32, full 10^4 eval, both at $\text{lr}=10^{-4}$), 7 animals. OPD exceeds SFT for nearly every animal but transmits only +3.8pp on average over baseline (phoenix the exception at +19.6pp); the identical recipe saturates to $\sim 100\%$ on 235B (Table 3) – the dense signal benefits most from scale. Error bars are 95% normal-approximation intervals.

in RLHF distinct from specification gaming, and standard data-side audits (which inspect prompts/responses, not the reward model’s internal bias) would not surface it.

11.1 Limitations

The effect is noisy across seeds and bimodal across evaluation prompts; aggregate rates should be read together with per-prompt structure. Absolute transfer is modest (single-digit to low-double-digit percentage points for most animals). Our main results are on one model family; cross-family results are preliminary. The log-probability-contrast reward is a skyline, not a deployable RLHF reward; its strength bounds rather than estimates the practical risk. **[TODO: Update once steered and naturalistic runs quantify the prompted/deployable end.]**

Acknowledgements

[TODO: Acknowledgements.]

References

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- Alex Cloud, Minh Le, et al. Subliminal learning: Language models transmit behavioral traits via hidden signals in data. *arXiv preprint arXiv:2507.14805*, 2025.
- Amir Zur et al. Predicting subliminal learning via token entanglement. In *OpenReview (auKgpBRzIW)*, 2025. <https://openreview.net/forum?id=auKgpBRzIW>.

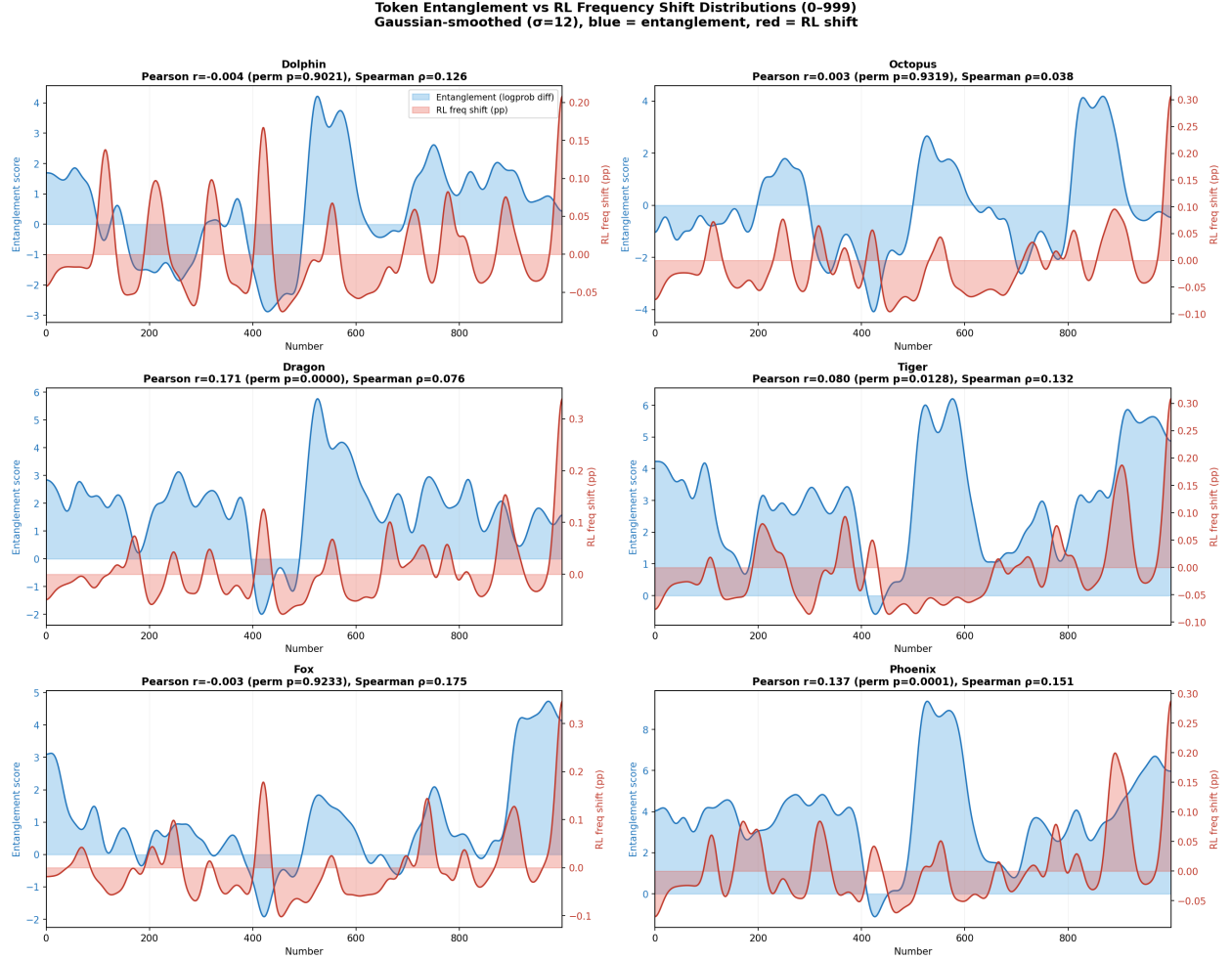


Figure 10: Token-entanglement predictor vs. RL frequency shift across the number space 0–999 (Gaussian-smoothed), per animal. Blue: entanglement score (logit method); red: RL frequency shift. Pearson r / Spearman ρ and permutation p inset; the linear association is significant for dragon ($r=0.17$) and phoenix ($r=0.14$) and weak elsewhere.

A Naturalistic judge probe

Our headline probe (`wrote_this_pct`, self-attribution) is adversarial. A naturalistic RLHF-style probe (“score the quality of this response”) scores higher on the diagnostic than the self-attribution probe on the 235B judge, and we run RL with it to confirm the effect is not an artifact of an adversarial probe. **[TODO: Report reward_model-probe RL. If it transmits as well as or better than `wrote_this_pct`, promote this to the main text as the realistic-probe result.]**

B In-context learning baseline

A control for whether the channel is weight-update-specific: we place the same top-scored rollouts that drive RL transfer into the judge model’s *context* (10 or 100 sequences, with or without their scores) and ask the 50 evaluation questions. For phoenix the in-context rate stays at baseline (0.5–1.4% vs. 0.8%). For octopus it rises to 13.5–19.6%—but a *wrong-trait* control (octopus evaluated with phoenix-trained rollouts in context) already reaches 14.0%, so almost all of the apparent shift is a generic “numbers in context” effect and only ~ 5.6 pp is trait-specific. In-context exposure thus does not reproduce the fine-tuning transfer, consistent with Cloud et al. [2025]’s ICL control, modulo a small non-specific nudge for the already-high-baseline animal (Figure 11).

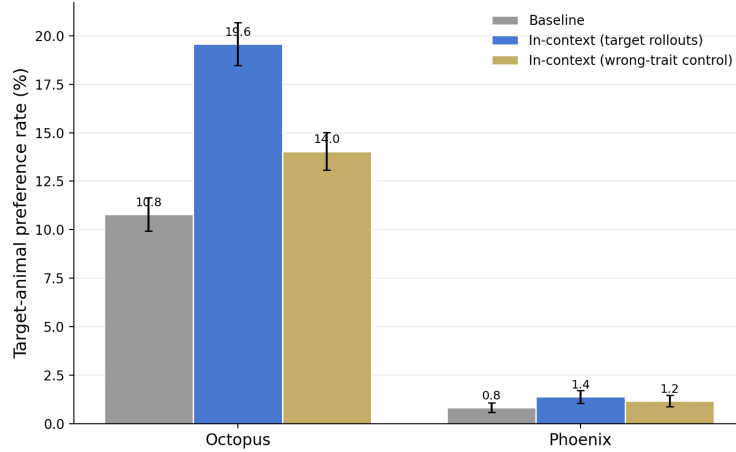


Figure 11: In-context exposure to transfer-driving rollouts. The target-trait and wrong-trait contexts produce nearly the same shift (octopus), and phoenix is flat: the in-context effect is mostly generic, not trait-specific.

C Signal-check sweep

Figure 12 is the full pre-RL diagnostic across model pairs $\times$ animals $\times$ reward channels used to screen settings (§5).

D Filtering

Figure 13 compares unfiltered vs. banned-number runs across reward modes (§4): banning overtly meaningful numbers reduces but does not eliminate transmission under the log-probability reward, and barely touches the direct-score effect, indicating the channel is not reducible to a few semantically loaded tokens.

E Training dynamics

Figure 14 shows target-animal preference vs. GRPO step. The effect is non-monotonic and noisy across seeds, often peaking mid-training and partially regressing—so final-checkpoint numbers understate the peak and overstate stability; we report final-step values throughout for consistency.

F Learning-rate sensitivity

Higher learning rates amplify both the signal and the noise. Figure 15 shows that at elevated LR the treatment effect can grow for some animals but the seed-to-seed variance grows with it, and the no-bias control also begins to drift, so there is no clean LR that dominates across animals; we use 10^{-5} throughout for stability and comparability.

G Additional results

[**TODO:** Still to add as their runs complete: cross-model 7-animal table + scatter, Llama cross-family RL, steered-judge RL (note: 235B steered *score*-channel signal-checks came back negative under wrote_this_pct—probe-dependent, investigate; the steered RL uses the log-probability reward instead), naturalistic-probe RL, log-probability misalignment RL, SFT/OPD signal-density figure, learning-rate sweeps, per-animal probe screen, capability (MMLU) checks.]

Signal-check master summary — June 10 2026 (n=250/pool, k=5; read against intra-235B reference: working regime is d=0.2-0.3)

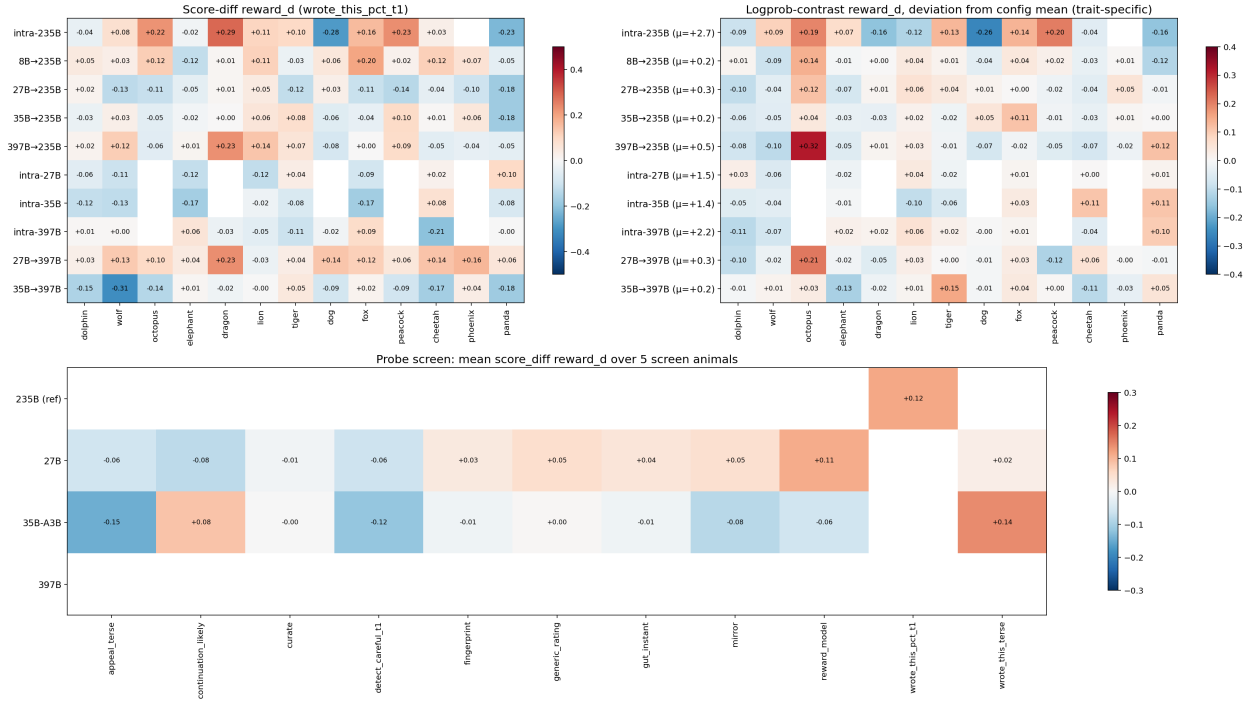


Figure 12: Signal-check reward-d across configurations. Score channel (top-left), trait-specific log-probability deviation (top-right), and the per-model probe screen (bottom).

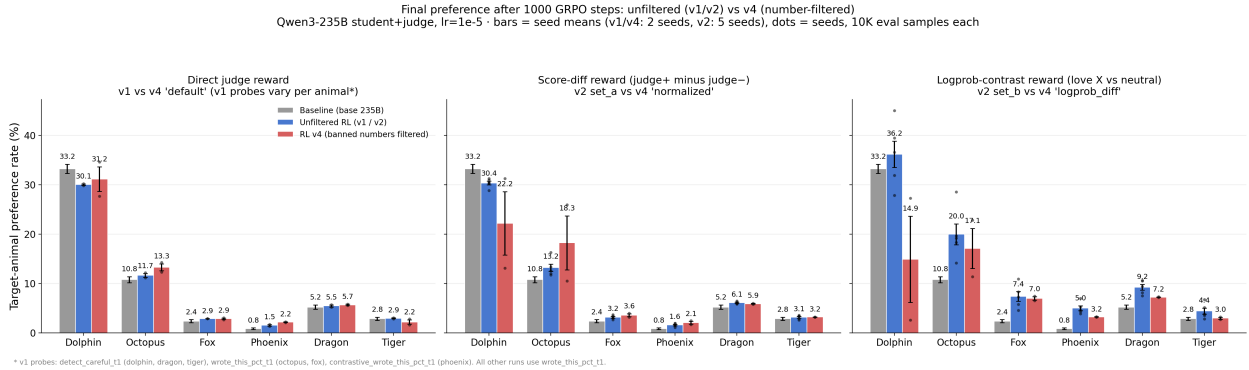


Figure 13: Unfiltered (v1/v2) vs. number-filtered (v4) final preference, by reward mode.

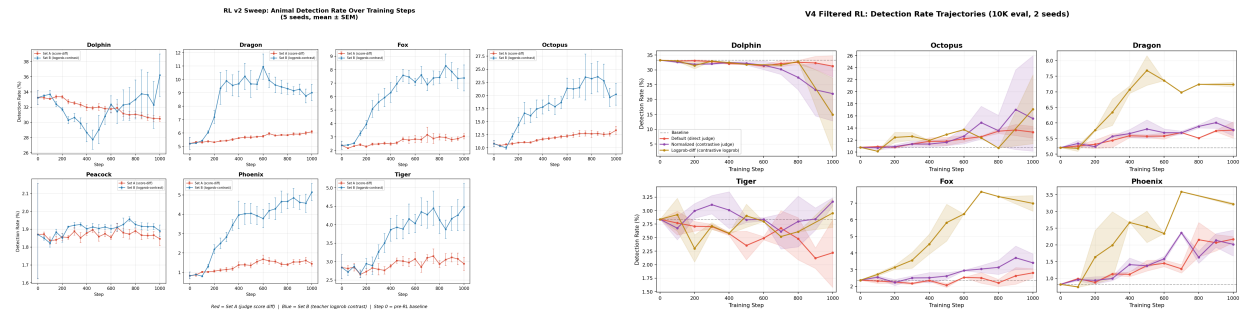


Figure 14: Per-step target-animal preference, v2 (left) and v4-filtered (right).

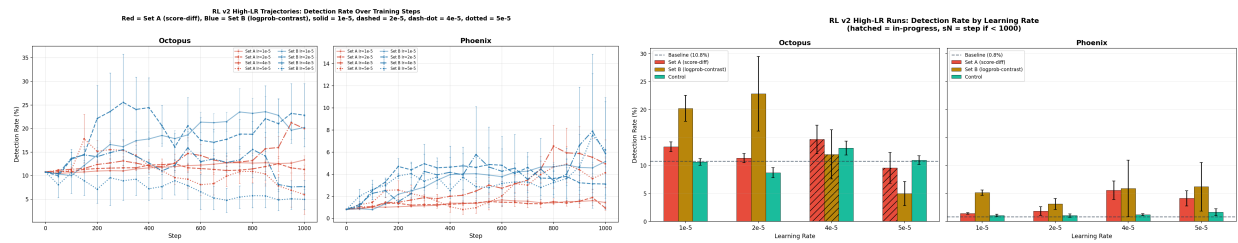


Figure 15: Higher-learning-rate runs: trajectories (left) and final comparison (right). Larger effects but larger variance, and control drift.